Prayas JEE AIR (2025)

WEEKLY TEST 20

DURATION : 180 Minutes | **DATE : 15/12/2024** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ **Invigilator's Signature** _____

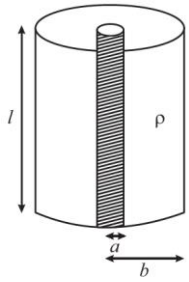
SECTION-I (PHYSICS)

Single Correct Type Questions

1. A person pushes a box on a rough horizontal platform surface. He applies a force of 200 N over a distance of 15 m. Thereafter, he gets progressively tired and his applied force reduces linearly with distance to 100 N. The total distance through which the box has been moved is 30m. What is the work done by the person during the total movement of the box?

(1) 5690 J (2) 3280 J
(3) 5250 J (4) 2780 J

2. Model a torch battery of length *l* to be made up of a thin cylindrical bar of radius '*a*' and a concentric thin cylindrical shell of radius '*b*' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value *R*, the maximum Joule heating in *R* will take place for



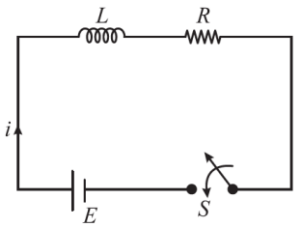
(1) $R=\frac{\rho}{\pi l}\ln\left(\frac{b}{a}\right)$ (2) $R=\frac{2\rho}{\pi l}\ln\left(\frac{b}{a}\right)$

(3) $R=\frac{\rho}{2\pi l}\ln\left(\frac{b}{a}\right)$ (4) $R=\frac{\rho}{2\pi l}\ln\left(\frac{a}{b}\right)$

3. The quantities $x=\frac{1}{\sqrt{\mu_0\varepsilon_0}}$, $y=\frac{E}{B}$ and $z=\frac{I}{CR}$ are defined where defined where *C*-capacitance, *R*-resistance, *l*-length, *E*-electric field, *B*-magnetic field and ε_0, μ_0-free space permittivity and permeability respectively. Then

(1) Only *x* and *y* have the same dimension.
(2) Only *x* and *z* have the same dimension
(3) *x*, *y* and *z* have the same dimension.
(4) Only *y* and *z* have the same dimension.

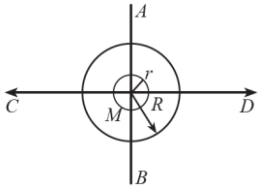
4. Consider the *LR* circuit shown in the figure. If the switch *S* is closed at $t = 0$ then the amount of charge that passes through the battery between $t=0$ $t=\frac{L}{R}$ is



(1) $\frac{EL}{7.3R^2}$ (2) $\frac{EL}{2.7R^2}$

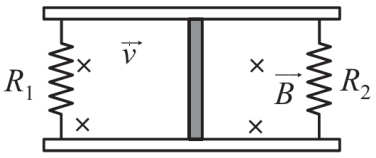
(3) $\frac{7.3EL}{R^2}$ (4) $\frac{2.7EL}{R^2}$

5. The figure shows two solid discs with radius *R* and *r* respectively. If mass per unit area is same for both, what is the ratio of *MI* of bigger disc around axis *AB* (which is ⊥ to the plane of the disc and passing through its centre) to *MI* of smaller disc around one of its diameters lying on its plane? Given '*M*' is the mass of the larger disc. (MI stands for moment of inertia)



(1) $R^2 : r^2$ (2) $4r^4 : R^4$
(3) $2R^4 : r^4$ (4) $2R^2 : r^2$

6. A conducting bar of length *L* is free to slide on two parallel conducting rails as shown in the figure.



Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ pointing into the page. An external agent pulls the bar to the left at a constant speed *v*. The correct statement about the directions of induced currents I_1 and I_2 flowing through R_1 and R_2 respectively is

(1) I_1 is in anticlockwise direction and I_2 is in clockwise direction
(2) Both I_1 and I_2 are in anticlockwise direction
(3) Both I_1 and I_2 are in clockwise direction
(4) I_1 is in clockwise direction and I_2 is in anticlockwise direction

7. If the potential energy between two molecules is given by $U=-\frac{A}{r^6}+\frac{B}{r^{12}}$, then at equilibrium, separation between molecules, and the potential energy are

(1) $\left(\frac{2B}{A}\right)^{1/6},\frac{A^2}{2B}$ (2) $\left(\frac{B}{2A}\right)^{1/6},-\frac{A^2}{2B}$

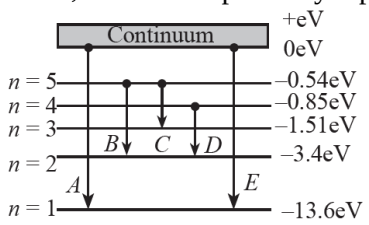
(3) $\left(\frac{2B}{A}\right)^{1/6},-\frac{A^2}{4B}$ (4) $\left(\frac{B}{A}\right)^{1/6},0$

8. A uniform cylindrical rod of length L and radius r, is made from a material whose Young's modulus of Elasticity equals Y. When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F, its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is equals to:

(1) $\frac{9F}{(\pi r^2 YT)}$ (2) $\frac{F}{(3\pi r^2 YT)}$

(3) $\frac{3F}{(\pi r^2 YT)}$ (4) $\frac{6F}{(\pi r^2 YT)}$

9. In the given, figure, the energy levels of hydrogen atom have been shown along with some transitions marked A, B, C, D and E. The transitions A, B and C respectively represent:



(1) The series limit of Lyman series, second member of Balmer series and second member of Paschen series.

(2) The first member of the Lyman series, third member of Balmer series and second member of Paschen series.

(3) The ionization potential of hydrogen, second member of Balmer series and third member of Paschen series.

(4) The series limit of Lyman series, third member of Balmer series and second member of Paschen series.

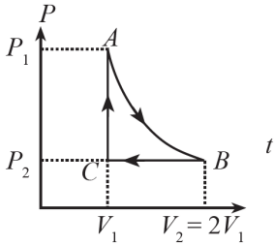
10. n moles of perfect gas undergoes a cyclic process $ABCA$ (see figure) consisting of the following process.

$A \rightarrow B$: Isothermal expansion at temperature T so that the volume is doubled from V_1 to $V_2 = 2V_1$ and Pressure changes from P_1 to P_2.

$B \rightarrow C$: Isobaric compression at pressure P_2 to initial volume V_1.

$C \rightarrow A$: Isochoric change leading to change of pressure from P_2 to P_1.

Total work done in the complete cycle $ABCA$ is:



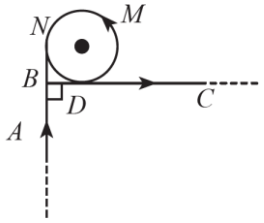
(1) $nRT\left(\ln 2 + \frac{1}{2}\right)$

(2) $nRT\left(\ln 2 - \frac{1}{2}\right)$

(3) 0

(4) $nRT \ln 2$

11. A very long wire $ABADMNDC$ is shown in figure carrying current I. AB and BC parts are straight, long and at right angle. At D wire forms a circular turn $DMND$ of radius R. AB, BC parts are tangential to circular turn at N and D. Magnetic field at the centre of circle is:



(1) $\frac{\mu_0 I}{2R}$ (2) $\frac{\mu_0 I}{2\pi R}\left(\pi - \frac{1}{\sqrt{2}}\right)$

(3) $\frac{\mu_0 I}{2\pi R}(\pi + 1)$ (4) $\frac{\mu_0 I}{2\pi R}\left(\pi + \frac{1}{\sqrt{2}}\right)$

12. Given below are two statements:

Statement-I: Acceleration due to earth's gravity decreases as you go 'up' or 'down' from earth's surface.

Statement-II: Acceleration due to earth's gravity is same at a height 'h' and depth 'd' from earth's surface, if $h = d$.

In the light of above statements, choose the most appropriate answer from the options given below

(1) Statement-I is incorrect but Statement-II is correct

(2) Both Statement-I and Statement-II are incorrect

(3) Statement-I is correct but Statement-II is incorrect

(4) Both Statement-I and II are correct

13. Let a total charge $2Q$ be distributed in a sphere of radius R, with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B, of $-Q$ each are placed on diametrically opposite points, at equal distance, a, from the centre. If A and B do not experience any force, then

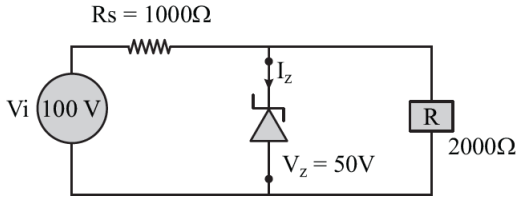
(1) $a = \frac{3R}{2^{1/4}}$ (2) $a = \frac{R}{\sqrt{3}}$

(3) $a = 8^{-1/4} R$ (4) $a = 2^{-1/4} R$

14. In a resonance tube experiment when the tube is filled with water up to a height of 17.0 cm from bottom, it resonates with a given tuning fork. When the water level is raised the next resonance with the same tuning fork occurs at a height of 24.5 cm. If the velocity of sound in air is 330 m/s, the tuning fork frequency is

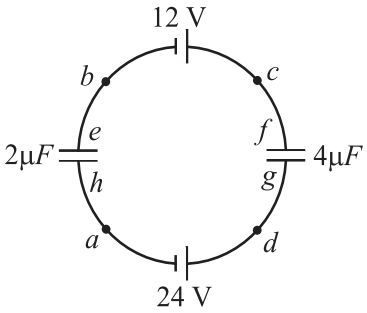
(1) 2200 Hz (2) 3300 Hz
(3) 1100 Hz (4) 550 Hz

15. For the circuit shown below, calculate the value of I_z:



(1) 0.15 A (2) 0.1 A
(3) 25 mA (4) 0.05 A

16. With reference to the circuit diagram shown. Match the following



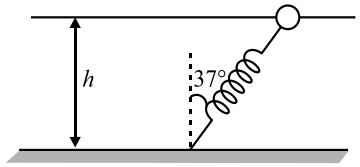
Column-I		Column-II	
A	$v_d - v_c$ (in volt)	P	–8
B	$v_a - v_b$ (in volt)	Q	+4
C	Work done by 12 V cell is (in μJ)	R	–192
D	Work done by 24 V cell is (in μJ)	S	384

(1) A→P; B→Q; C→R; D→S
(2) A→P; B→Q; C→S; D→R
(3) A→Q; B→P; C→S; D→R
(4) A→Q; B→P; C→R; D→S

17. A metal surface is illuminated by a radiation of wavelength 4500Å. The ejected photoelectron enters a constant magnetic field of 2 mT making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius 2mm, the work function of the metal is approximately:

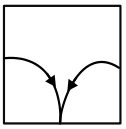
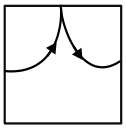
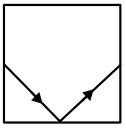
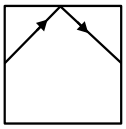
(1) 1.36 eV
(2) 1.69 eV
(3) 2.7 eV
(4) 2.23 eV

18. One end of a spring of natural length h and spring constant k is fixed at the ground and the other is fitted with a smooth ring of mass m which is allowed to slide on a horizontal rod fixed at a height h. Initially the spring makes an angle of 37° with the vertical when the system is released from rest. The speed of the ring when the spring becomes vertical is



(1) $\sqrt{\frac{k}{m}}$ (2) $\frac{h}{2}\sqrt{\frac{k}{m}}$
(3) $\frac{h}{3}\sqrt{\frac{k}{m}}$ (4) $\frac{h}{4}\sqrt{\frac{k}{m}}$

19. A cubic container is filled with a liquid whose refractive index increases linearly from top to bottom. Which of the following represents the path of a ray of light inside the liquid?

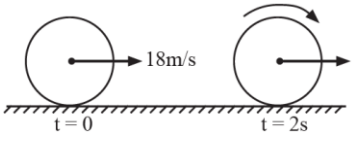
(1) 
(2) 
(3) 
(4) 

20. A ball of mass 200 g rests on a vertical post of height 20 m. A bullet of mass 10 g, travelling in horizontal direction, hits the centre of the ball. After collision both travels independently. The ball hits the ground at a distance 30 m and the bullet at a distance of 120 m from the foot of the post. The value of initial velocity of the bullet will be (if g = 10 m/s^2):

(1) 120 m/s (2) 60 m/s
(3) 400 m/s (4) 360 m/s

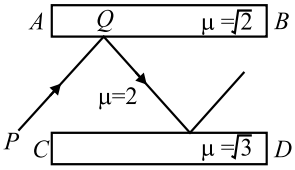
Integer Type Questions

21. A uniform disc of mass 0.5 kg and radius r is projected with velocity 18 m/s at t = 0 s on a rough horizontal surface. It starts off with a purely sliding motion at t=0s. After 2 s it acquires a purely rolling motion (see figure). The total kinetic energy of the disc after 2 s will be _____J. (given, coefficient of friction is 0.3 and g = 10 m/s^2)

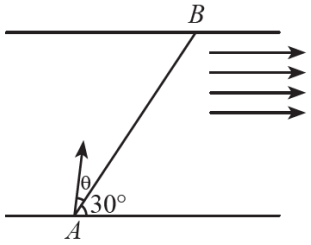


22. Two light waves of wavelengths 800 nm and 600 nm are used in Young's double slit experiment to obtain interference fringes on a screen placed 7 m away from the plane of slits. If the two slits are separated by 0.35 mm, then shortest distance from the central bright maximum to the point where the bright fringes of the two wavelength coincide will be _____ mm.

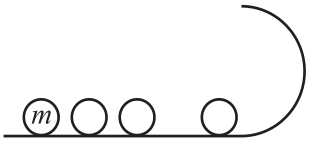
23. *AB* and *CD* are two slabs. The medium between the slabs has refractive index 2. The minimum angle of incidence at *Q*, so that the ray is totally reflected by both the slabs is *n* degrees. Find the value of *n*



24. A swimmer wants to cross a river from point *A* to point *B*. Line *AB* makes an angle of 30° with the flow of river. Magnitude of velocity of the swimmer in still water is same as that of the river. The angle θ with the line *AB* should be _____°, so that the swimmer reaches point *B*.



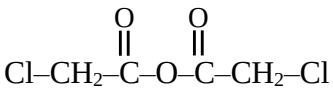
25. *n* elastic balls are placed at rest on a smooth horizontal plane which is circular at the ends with radius *r* as shown in the figure. The masses of the balls are $m, \frac{m}{2}, \frac{m}{2^2}.....\frac{m}{2^{n-1}}$ respectively. The minimum velocity which should be imparted to the first ball of mass *m* such that n^{th} ball completes the vertical circle is $\left(\frac{a}{b}\right)^{n-1}\sqrt{5gr}$. Find the value of $a + b$. (a & b are coprime integers)



SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. Given below is an organic compounds:



IUPAC name is correctly represented by which of the following option:

(1) Bis (Chloroacetic anhydride)
(2) Bis (Chloroethanoic anhydride)
(3) Chloroethanoic anhydride
(4) 1-Chloroethanoyloxy-2-chloroethanone

27. Oxygen as well as nitrogen exist as diatomic molecules. However, the subsequent elements can exist as S_8 and P_4 molecules respectively, because:

(1) Phosphorus and sulphur are solids.
(2) Phosphorus and sulphur catenate due to the existence of d-orbitals and form structures having less strain.
(3) Phosphorus and sulphur polymerise as soon as they are formed.
(4) Catenation tendency of P and S is stronger because of the high P – P and S – S bond energies as compared to N – N and O – O bond energies.

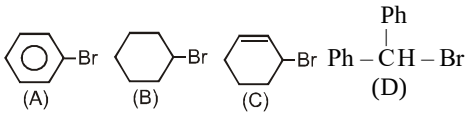
28. Among the statements given in the options, the incorrect statement is

(1) First ionisation energies of elements along a period do not vary in a regular manner with the increase in atomic number.
(2) Non–metallic elements are lesser in number as compared to metallic elements.
(3) The properties of elements are periodic function of their atomic numbers as per Modern periodic law.
(4) For transition elements, the d–subshells are filled with electrons monotonically with increase in atomic number.

29. When some amount of solute is added to 800 g of H_2O (solvent), mole fraction of solute is changed from $\frac{1}{4}$ to $\frac{1}{2}$. The ratio of molality of initial solution before addition and final solution after addition of solute will be:

(1) 1 : 3 (2) 1 : 2
(3) 2 : 3 (4) 1 : 4

30. Correct increasing order of rate of hydrolysis of following compounds is:



(1) A < B < C < D (2) D < C < B < A
(3) A < B < D < C (4) A < D < B < C

31. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{PBr_3} A \xrightarrow{Mg/dry\ ether} B$

$\xrightarrow{\text{O}\ /\ CH_2-CH_2} C \xrightarrow{H_2O} D$

D is:

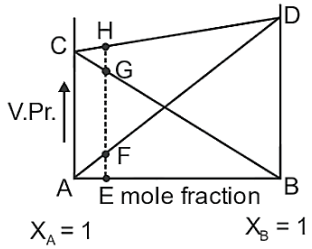
(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-O-CH_2-CH_3$

(2) $CH_3-O-\underset{\underset{CH_3}{|}}{CH}-CH_2CH_3$

(3) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2CH_2OH$

(4) $CH_3-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2OH$

32. Based on the given diagram, which of the following statements regarding the homogeneous solutions of two volatile liquids are correct?



(I) Plots AD and BC show that Raoult's law is obeyed for the solution in which B is a solvent and A is the solute and as well as for that in which A is solvent and B is solute.

(II) Plot CD is also followed by a non-ideal solution.

(III) EF + EG = EH; and AC and BD correspond to the vapour pressures of the pure solvents A and B respectively.

Select the correct answer using the options given below

(1) Only I

(2) Only II and III

(3) Only I and III

(4) All of these

33. Consider $n_1 + n_2 = 7$ & $n_2 - n_1 = 3$ where n_1 and n_2 correspond to the boundary value of principal quantum numbers of a portion of emission spectrum of H atom. The longest wavelength possible corresponding to the last line as per these boundary values is:

(Given $R_H = 1.097 \times 10^7\ m^{-1}$)

(1) 9×10^{-6}

(2) 4×10^{-6}

(3) 15×10^{-7}

(4) 10×10^{-7}

34. **Statement-1:** $[Fe(CN)_6]^{3-}$ is weakly paramagnetic while $[Fe(CN)_6]^{4-}$ is diamagnetic **and**

Statement-2: Both $[Fe(CN)_6]^{3-}$ and $[Fe(CN)_6]^{4-}$ involve d^2sp^3 hybridisation. Due to high field strength of $CN^{\ominus}$ ligand only one of the d-orbital is singly occupied in $[Fe(CN)_6]^{3-}$ and all the t_{2g} orbitals are doubly occupied in $[Fe(CN)_6]^{4-}$.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(3) Statement-1 is True, Statement -2 is False.

(4) Statement-1 is False, Statement-2 is True.

35. Which of the following species is not correctly matched with the shape?

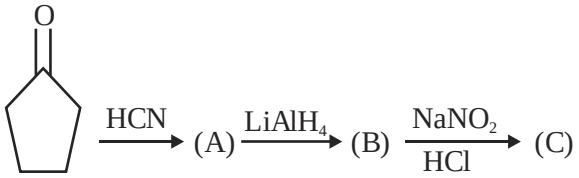
(1) N_3^- – Linear

(2) SF_4 – See Saw

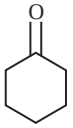
(3) ClO_3^- – Pyramidal

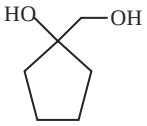
(4) PCl_6^- – Square pyramidal

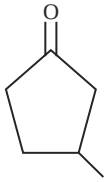
36. Consider the following reaction sequence.

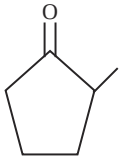


Find product C in the above reaction sequence is:

(1) 

(2) 

(3) 

(4) 

37. In the detection of nitrogen of an organic compound by Lassaigne's test, Prussian blue colour is obtained. This is due to the formation of which of the following complexes?

(1) $Fe_2\left[Fe(CN)_6\right]$ (2) $Fe_4\left[Fe(CN)_6\right]_3$

(3) $Fe_3\left[Fe(CN)_6\right]_4$ (4) $Na_4\left[Fe(CN)_6\right]$

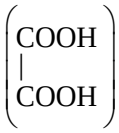
38. The correct increasing order of O — O bond length in $O_2[PtF_6]$, O_2, BaO_2 and KO_2 are

(1) $O_2[PtF_6] < O_2 < KO_2 < BaO_2$

(2) $O_2 < O_2[PtF_6] < KO_2 < BaO_2$

(3) $O_2[PtF_6] < O_2 < BaO_2 < KO_2$

(4) $O_2[PtF_6] < KO_2 < O_2 < BaO_2$

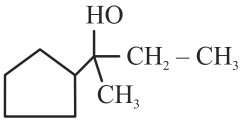
39. Compound among following produces only oxalic acid  as major product on reaction with O_3 / H_2O_2.

(1) 

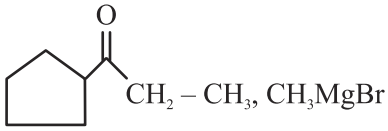
(2) 

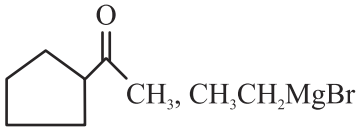
(3) $H_2C = CH_2$

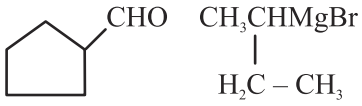
(4) 

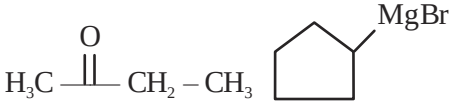
40. To synthesize 

Which of the following reactants cannot be suitable:

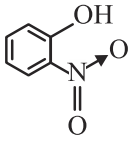
(1) 

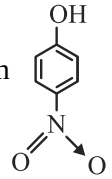
(2) 

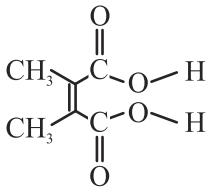
(3) 

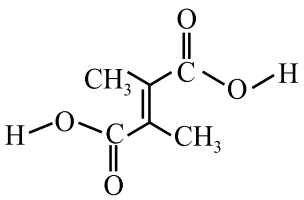
(4) 

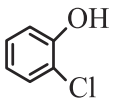
41. Which of the following statement is false?

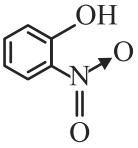
(1) 

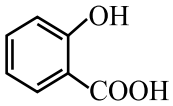
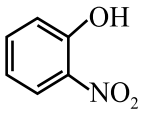
has lower boiling point than 

(2) 

is less stable than 

(3) 

has greater pKa than 

(4)  has greater pka value than 

42. Solubility product (K_{sp}) of cupric iodate $[Cu(IO_3)_2]$ is 7.4×10^{-8}. In which of the following mixtures, a ppt of cupric iodate will be formed?

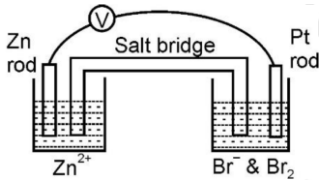
(1) Equal volume of 0.002 M $NaIO_3$ and 0.002 M $Cu(ClO_3)_2$

(2) Equal volume of 0.001 M $NaIO_3$ and 0.002 M $Cu(ClO_3)_2$

(3) Equal volume of 0.004 M $NaIO_3$ and 0.002 M $Cu(ClO_3)_2$

(4) Equal volume of 0.02 M $NaIO_3$ and 0.002 M $Cu(ClO_3)_2$

43. Consider the following diagram representing a working galvanic cell



Which of the following decreases during the working of cell?

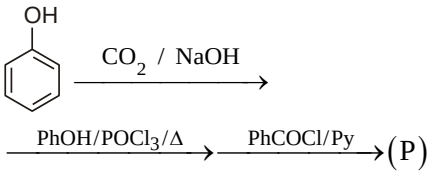
(1) $[Zn^{2+}]$, $[Br_2]$ & $[Br^-]$

(2) $[Br_2]$, mass of Zn rod, Mass of Pt rod

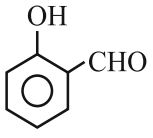
(3) $[Br_2]$, mass of Zn rod

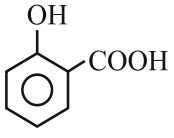
(4) $[Br_2]$, $[Br^-]$, mass of Pt rod

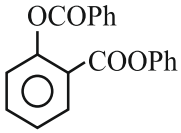
44. Consider the following reaction sequence:

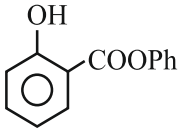


The final product (P) in the given reaction sequence is:

(1) 

(2) 

(3) 

(4) 

45. Orange solid (S_1) $\xrightarrow{\Delta}$ diamagnetic gas + only green residue

Orange solid (S_2) $\xrightarrow{\Delta}$ paramagnetic gas + green solid + yellow solid

S_1 and S_2 are respectively:

(1) $(NH_4)_2Cr_2O_7$, KNO_3

(2) $Na_2Cr_2O_7$, $K_2Cr_2O_7$

(3) $K_2Cr_2O_7$, $(NH_4)_2Cr_2O_7$

(4) $(NH_4)_2Cr_2O_7$, $K_2Cr_2O_7$

Integer Type Questions

46. A 140 g sample of nitrogen gas was initially at 50 atm and 27°C. The magnitude of work done, if the gas is expanded reversibly and isothermally to 1 atm, is 'x' kcal. Then find the value of x in nearest integer. (log 5 = 0.7)

47. From the given list, the number of complexes showing cis-trans isomerism is_____.

$CoCl_3.4NH_3$, $CoCl_3.5NH_3$, $CoCl_3.6NH_3$, $CoCl(NO_3)_2.5NH_3$

48. Consider the following equilibrium at 27°C:

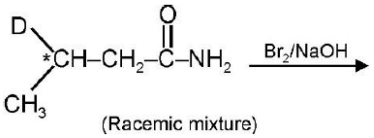
$Br_2(l) + Cl_2(g) \rightleftharpoons 2BrCl(g)$; $K_P = 1$atm

The reaction starts with $Br_2(l)$ and 1 mole of $Cl_2(g)$ in a closed container. The equilibrium partial pressure of $Cl_2(g)$ is 0.01 atm. If the minimum mass of $Br_2(l)$ initially taken is a g then the value of $\sqrt{a}$ is _____ (vapour pressure of $Br_2(l)$ at given temperature is 0.1 atm) (Atomic mass of Br = 80 u)

49. H_2O_2 decomposes to give H_2O and O_2. The decomposition reaction is a first order reaction. Consider a sample of H_2O_2 labelled as 20 volumes, which was left open. As a result of this, some H_2O_2 decomposed. After 6 hours, 10 mL of this solution was diluted to 100 mL to determine the new volume strength. 10 mL of the diluted solution was now titrated against 25 mL of 0.025 M $KMnO_4$ in acidic medium. If it is given that the rate constant for the decomposition of H_2O_2 in hr^{-1} is k, then 100 k is: (Nearest integer)

$$\left[\text{Given}: \log\left(\frac{1.786}{1.563}\right) = 0.0579\right]$$

50. Consider the following reaction



(Racemic mixture)

How many different amines are obtained at the end of the reaction?

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. The domain of the function

$$f(x) = \sqrt{\frac{(\sin x - e^x)(\cos x - 3)(\ln x + 2)}{\left(x^4 - x + \frac{1}{2}\right)(2x-7)^{30}}} \text{ belongs to}$$

(1) $\left[\frac{1}{e^2}, \frac{7}{2}\right) \cup \left(\frac{7}{2}, \infty\right)$

(2) $\left(-\frac{1}{2}, 0\right)$

(3) $\left(0, \frac{1}{e^5}\right)$

(4) ϕ

52. Let $J = \int_{-5}^{-4} (3-x^2)\tan(3-x^2)dx$ and

$K = \int_{-2}^{-1} (6-6x+x^2)\tan(6x-x^2-6)dx$, then

(J + K) equals

(1) 1

(2) 0

(3) 3

(4) 4

53. If $\sin^{-1}\left(\frac{5}{x}\right) + \sin^{-1}\left(\frac{12}{x}\right) = \sin^{-1}\left(\frac{2}{x}\right) + \cos^{-1}\left(\frac{2}{x}\right)$

then the value of x is equal to

(1) 21 (2) 13

(3) 12 (4) 10

54. If the asymptote of the hyperbola $(x + y + 1)^2 - (x - y - 3)^2 = 5$ cuts each other at A and the co-ordinate axes at B and C, then the radius of circle passing through the points A, B, C is:

(1) 3 (2) $\frac{\sqrt{5}}{2}$

(3) $\frac{\sqrt{3}}{2}$ (4) $\sqrt{3}$

55. Let T be the term independent of x in the expansion of $(x^{-1}\,\text{cosec}\phi + x\,\sec\phi)^{16}$ and a = minimum value of T for $\phi \in \left[\frac{\pi}{8}, \frac{\pi}{4}\right]$ and b = minimum value of T for $\phi \in \left[\frac{\pi}{16}, \frac{\pi}{8}\right]$, then $(a + b)$ is equal to

(1) ${}^{16}C_4 \cdot 2^{12} \cdot 17$ (2) ${}^{16}C_8 \cdot 2^8 \cdot 17$

(3) ${}^{16}C_{12} \cdot 2^4 \cdot 13$ (4) ${}^{16}C_8 \cdot 2^{16} \cdot 13$

56. If A and B are square matrices of order 3 where $|A| = -2$ and $|B| = 1$, then $|(A^{-1})\ \text{adj}(B^{-1})\ \text{adj}(2A^{-1})|$ is equal to

(1) 8 (2) –8
(3) 1 (4) –1

57. Value of $\int \frac{9\cos 2x + 12\cos x - 6\sin x}{(2+3\sin x)(4+3\cos x)}dx$ is

(1) $\log_e\left(\frac{2+3\sin x}{4+3\cos x}\right)+c$

(2) $\log_e\left(\frac{4+3\cos x}{2+3\sin x}\right)+c$

(3) $\log_e((2+3\sin x)(4+3\cos x))+c$

(4) $-\log_e((2+3\sin x)(4+3\cos x))+c$

58. Sum of real solutions of equation

$\log_4(x^2-x) = \log_2\left(2^{\frac{1}{4}}\cdot 4^{\frac{1}{16}}\cdot 8^{\frac{1}{48}}\cdot 16^{\frac{1}{128}}\ldots\text{upto}\ \infty\right)$

is equal to

(1) –2 (2) 2
(3) 1 (4) –1

59. If the equation $x^3 - (p+1)x^2 + (q-p)x - q = 0$ has roots $\tan(\theta_r)$ where $r = 1, 2, 3$ and $q - p \neq 1$ where $p, q \in R$, then

(1) $\sin(\theta_1+\theta_2+\theta_3)=\frac{1}{2}$

(2) $\cot(\theta_1+\theta_2+\theta_3)=1$

(3) $\theta_1+\theta_2+\theta_3=\frac{3\pi}{4}$

(4) $\theta_1+\theta_2+\theta_3=\pi$

60. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $(A - B) \cup (B - A)$ is

(1) {1, 2, 3, 5} (2) {1, 2, 4}
(3) {1, 2, 3, 4} (4) {1, 2, 4, 5}

61. If a point z_1 satisfying $|z + 8| \leq 2$ and z_2 satisfying $|z + 3| + |z - 3| \leq 10$, then the range of $|z_1 - z_2|$ is

(1) [10, 15] (2) [1, 15]
(3) [2, 10] (4) [1, 10]

62. The sum of abscissa of the points of inflection of curve $y = f(x) = \frac{x^7}{42} - \frac{2x^6}{15} + \frac{3x^5}{20} + \frac{x^4}{3} - \frac{2x^3}{3} + 99x + 2010$ is,

(1) –1
(2) 0
(3) 2
(4) –2

63. The area of a plane figure bounded by the points of the lines $\max(x, y) = 1$ and $x^2 + y^2 = 1$ is

(1) $\frac{\pi}{2}$ sq.units (2) $\frac{\pi}{3}$ sq.units

(3) $1-\frac{\pi}{4}$ sq.units (4) $2 - \pi$

64. Let $AB : y = x$ be a chord to the parabola $y^2 + 4x = 4$ and P be any variable point on the ellipse $x^2 + 9(y - 4)^2 - 9 = 0$. If Δ_1 and Δ_2 be the minimum and maximum area of triangle ABP respectively then

(1) $\Delta_2 = 2\sqrt{2}(4+\sqrt{10})$

(2) $\Delta_2 = \Delta_1 = 8\sqrt{5}$

(3) $\frac{\Delta_2}{\Delta_1} = \frac{13-4\sqrt{10}}{3}$

(4) $\Delta_1 = \sqrt{2}(4-\sqrt{10})$

65. If $f(n) = \lim_{x\to 0}\left(\left(1+\sin\frac{x}{2}\right)\left(1+\sin\frac{x}{2^2}\right)....\left(1+\sin\frac{x}{2^n}\right)\right)^{\frac{1}{x}}$, then $\lim_{n\to\infty} f(n) =$

(1) 1 (2) e
(3) 0 (4) ∞

66. Let $\vec{a} = \hat{i} + \hat{j}, \vec{b} = \hat{j} + \hat{k}$ and $\vec{c} = \alpha\vec{a} + \beta\vec{b}$. If the vectors, $\hat{i} - 2\hat{j} + \hat{k}, 3\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c}$ are coplanar then $\frac{\alpha}{\beta}$ is

(1) 1 (2) 2
(3) 3 (4) –3

67. A function f is defined on [–3, 3] as $f(x) = \begin{cases} \min\{|x|, 2-x^2\}, & -2 \leq x \leq 2 \\ [|x|], & 2 < |x| \leq 3 \end{cases}$ where $[x]$ denotes the greatest integer $\leq x$. The number of points, where f is not differentiable in (–3, 3) is m and number of points where f is discontinuous in (–3, 3) is n, then $m + n$ is

(1) 7 (2) 5
(3) 6 (4) 8

68. Incentre of the triangle formed by the lines $y = x$, $y = 3x$ and $y = 8 - 3x$ is

(1) $\left(\frac{4}{4}, \frac{2-\sqrt{5}}{3}\right)$ (2) $\left(\frac{4}{4}, \frac{2+\sqrt{5}}{3}\right)$

(3) $\left(\frac{4}{3}, \frac{2+2\sqrt{5}}{3}\right)$ (4) $\left(\frac{4}{3}, \frac{2-2\sqrt{5}}{3}\right)$

69. A line L passing through the point $P(0, 1, -1)$, and is perpendicular to both the lines $\frac{x-2}{2} = \frac{y-4}{1} = \frac{z+2}{4}$ and $\frac{x+2}{3} = \frac{y+4}{2} = \frac{z-1}{-2}$. If the position vector of point Q on L is (a, b, c) such that $(PQ)^2 = 357$, then $a + 2b + 3c$ is equal to

(1) 26
(2) 24
(3) –24
(4) 7

70. If a singular matrix $A = [a_{ij}]_{2\times 2}$ always commute with $B = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$ such that $\frac{a_{11}}{a_{12}} = \sqrt{k}$, then k is

(1) 0
(2) 1
(3) 2
(4) 3

Integer Type Questions

71. Let the solution curve of the differential equation $xdy = \left(\sqrt{16x^2 + y^2} + y\right)dx$, $y(1) = 3$ be $y = y(x)$. Then $y(2)$ is equal to

72. In a game of chance, a player throws a pair of dice and scores points equal to the difference between the numbers on the two dice. Winner is the person who scores exactly 5 points more than his opponent. If two players are playing this game only one time, then the probability that neither of them wins, is $\frac{p}{q}$ (where p, q are co-prime), then $p + q$ is

73. The number of five-digit numbers that can be formed using its digits 1, 2, 3, 4, 5, 6, 7, 8, 9 in which one digit appears once and two digits appear twice (*e.g.*, 44533) is k then the value of $\frac{k}{1890}$ is equal to _____

74. If

$$y(\alpha) = \sqrt{2\left(\frac{\tan\alpha + \cot\alpha}{1 + \tan^2\alpha}\right) + \frac{1}{\sin^2\alpha}}, \alpha \in \left(\frac{3\pi}{4}, \pi\right),$$

then $\frac{dy}{d\alpha}$ at $\alpha = \frac{5\pi}{6}$ is

75. If the point on line $x + y + 13 = 0$, which is nearest to circle $x^2 + y^2 + 4x + 6y - 5 = 0$ is (p, q), then $|p + q|$ is



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if